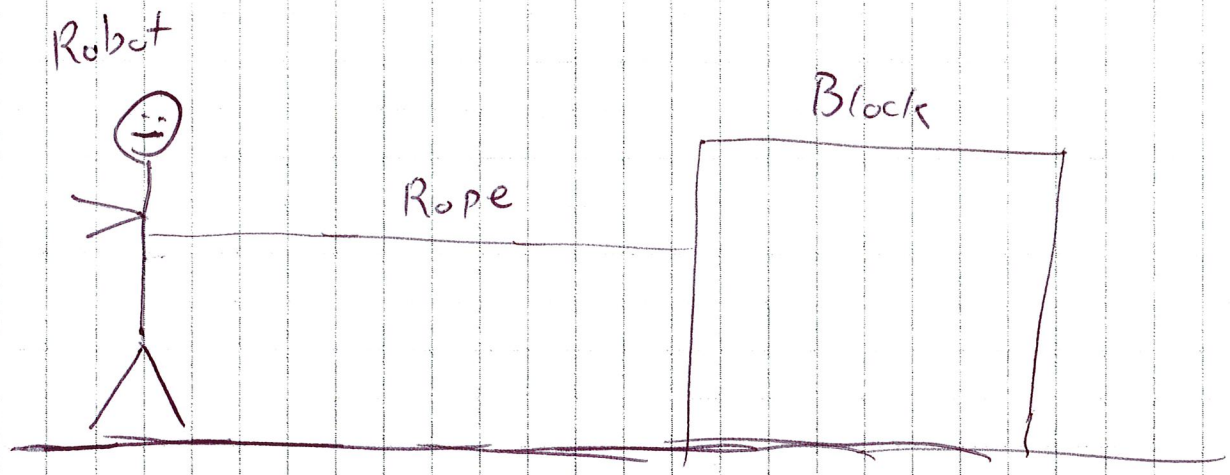
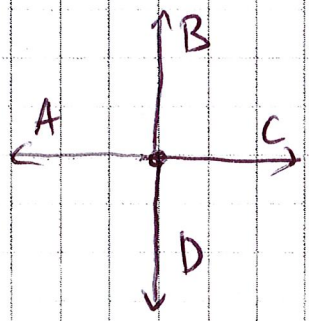
DIRECTIONS: Fill out this table, indicating the agency of each force, and each forces pair.

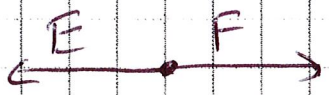
Force	**Type**	**Agent**	**Object**	**Newton's Third Law Pair**
A	Static Friction			
B				
C				
D				
E	Force of Tension			
F	Force of Tension			
G				
H				
I				
J				
K				
L		Robot		
M		Block		
N				
O		Robot		
P		Block		



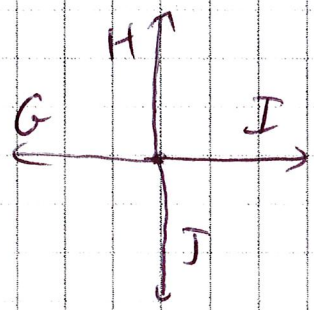
Free-Body Diagram of Robot



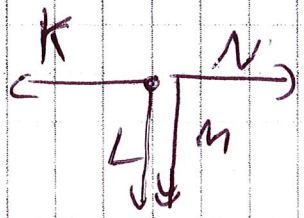
Free-Body Diagram of Rope



Free-Body Diagram of Block



Limited Free-Body Diagram of Floor



Limited Free-Body diagram of planet

- A robot drags a block on the ground by a rope
- Assume the rope is massless (not affected by gravity)
- There is friction on the ground
- Assume static friction is the force pushing the robot forwards
- Assume $\Sigma F = 0$ on the robot, rope, and block
- The rope pulling the rope can be called 'force of Tension'